

29/01/2024

Code No.: 22CSC01

CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.E. & B.Tech I Sem (Main & Backlog) Examination January 2024

Problem Solving and Programming

(Common to all branches)

Time: 3 Hours

Max Marks: 60

Note: Answer ALL questions from Part-A at one place in the same order and Part-B (Internal Choice)

Part - A

(5Q X 2M = 10 Marks)

	M	CO	BT
1 What is a flow chart? How it is different from algorithm.	(2)	1	1
2 Compare between break and continue with example.	(2)	2	2
3 List the different types of user defined data types.	(2)	2	1
4 Define Function. What is the advantage of modularizing a program into a function?	(2)	2	1
5 What is a file? How can we operate a file?	(2)	1	1

Part - B

(5Q X 10M = 50 Marks)

	M	CO	BT
6 (a) Discuss generation of programming languages. Is python a good choice of programming language for problem solving, Justify?	(5)	1	1
(b) Draw a flow chart for reversing a given number.	(5)	1	1
(OR)			
7 (a) With a neat block diagram of computer explain its components.	(5)	1	2
(b) Define algorithm. What are the different steps followed in the program development.	(5)	1	1
8 (a) Differentiate between list, tuple and set datatypes in python.	(6)	2	2
(b) Write a program to find whether a number is perfect or not. (A perfect no. is a no in which the sum of its proper divisors is equal to the no itself. Proper divisors of a no. are all divisors of a no. except itself, Example:6)	(4)	3	3
(OR)			
9 (a) Define a string. Explain the string handling functions with examples.	(6)	1	1
(b) Differentiate between local variables and global variables with example program.	(4)	2	2
10 (a) When we use nested loops. Explain with an example.	(5)	2,3	2
(b) Write a Python program to find largest and smallest number in a list.	(5)	2	3
(OR)			
11 (a) Explain selective or decision control statements with example.	(5)	2	2
(b) Write a Python program to print the lower triangle of a given 3*3 matrix.	(5)	3	3
12 (a) What is a dictionary? Specify the operations that we can perform on a dictionary.	(5)	5	1
(b) Write a function to search for a given element using linear search technique.	(5)	1	3

(OR)

- 13 (a) What is recursion? Write a program to print Fibonacci series using recursion. (5) 2,4 1
- (b) Write a bubble sort program by passing array to function. (5) 1 1
- 14 (a) Explain file handling in Python. (5) 2 2
- (b) Write a Python program to copy the contents of one file to another file. (5) 3 3
- (OR)
- 15 (a) Write a Python program to read a text file and convert the contents into upper case and print them. (5) 3 1
- (b) What is Cyclic References in garbage collection and how it is different from reference count method explain with an example. (5) 2 1
